Back to entropy

Choosing a base model by how well it reads your text

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Which base model will fine-tune best on our data?

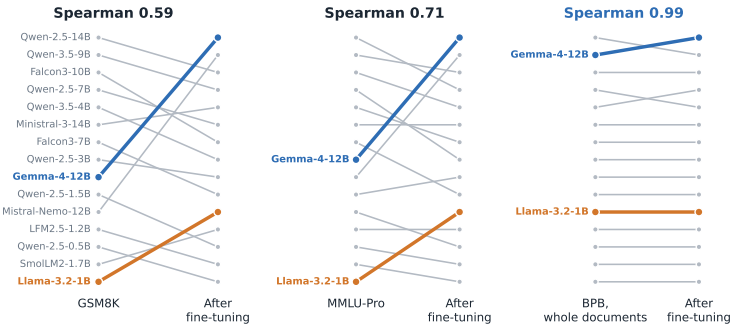
Today the choice is made from a scorecard.

- Fine-tuning every candidate would settle it, but needs a task, labels and a training run each.
- So we read MMLU-Pro, GSM8K, HellaSwag.
- Those grade a model **before** the fine-tuning it will receive.

Llama-3.2-1B vs **Qwen2.5-1.5B**	
HellaSwag	2.1 points apart
BPB on news documents	2.3% apart
GSM8K	55.3 **points apart**
Fine-tuned on news	not separated

Read the domain's documents; the fine-tuned order follows

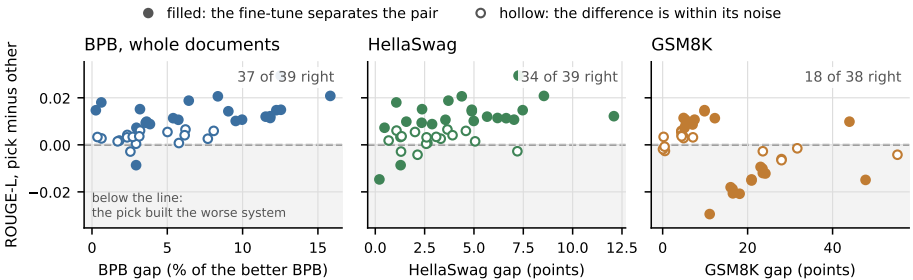
Fifteen base models, each fine-tuned on the same news-writing task



Bits per byte on whole news documents, no training: every model within **one rank** of its finish. Gemma-4-12B, 9th on GSM8K, builds the best system.

Pair by pair

39 pairs of models within 2× in size: which one fine-tunes better?



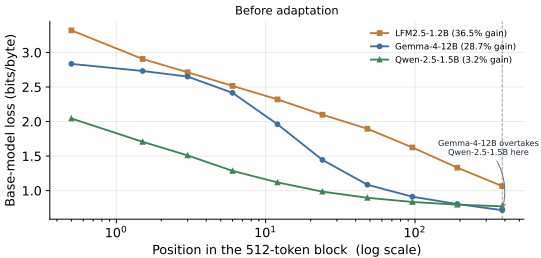
Document BPB **37 of 39** (21 of the 22 pairs the fine-tune resolves) HellaSwag 34 size 28 MMLU-Pro 20 the models' own untrained score on the task **20** GSM8K **18 of 38**. Same under a second, very different fine-tuning recipe.

Why this is not already how it is done

- The closest prior test (50 models of 1B): perplexity on a general held-out set orders fine-tuned pairs **worse than chance**; few-shot accuracy does best.
- Two details decide it, and the usual practice gets both wrong:
  1. **Which text**: it has to be the task's domain.
  2. **How it is cut**: whole documents, each from its own start, not fixed blocks from a concatenated stream.

Detail 1: blocks cut from a stream misrank strong models

The same news text, scored on 512-token blocks: 30 of 39 pairs



- Most blocks start mid-document, with no beginning-of-document token.
- Gemma-4-12B opens a block at 2.83 BPB against a 1.5B model's 2.04, and passes it by the end.
- Blocks: **13th of 15**. Whole documents: **2nd**. Fine-tuned: 1st.

Detail 2: the text has to be like the task's

Predicting the news fine-tune

- News documents: **37 of 39** pairs.
- arXiv maths documents: **23 of 39**, below ranking by size.

Agreement with benchmarks follows the text

	GSM8K	MMLU-Pro	HellaSwag
News	0.755	0.827	0.991
Reddit	0.691	0.764	0.973
Hacker News	0.782	0.873	0.982
Math (arXiv)	0.918	0.945	0.836

Eleven models, Spearman. HellaSwag tracks general prose; mathematics tracks GSM8K and MMLU-Pro. The rule was written down before the three new corpora were read.

No training, and no noise

Adapting first changes little

- A short LoRA adaptation lowers every model's loss by 2.0–7.9% on news.
- The document ranking barely moves: Spearman **0.993**, 38 of 39 pairs instead of 37.
- It does repair the block reading; that is how we found the problem.

Deterministic

- Re-running the reading reproduces every model's BPB exactly.
- Same weights, new host: GSM8K moves -0.60 **to** $+0.69$ points; one shipped decoding setting moves Gemma-4-12B's MMLU-Pro by **14.30**.

Why loss is the right quantity

- A base model acquires its capabilities by minimising cross-entropy on text; fine-tuning minimises it again on the task.
- Cross-entropy is a proper scoring rule. Over the domain's text p:

$$H(p, q) = \underbrace{H(p)}_{\text{same for every model}} + \mathrm{KL}(p \,\|\, q)$$

- Per **byte**, so every tokenizer is on one scale; on text **dated after every release**, so no candidate has trained on it.

Using it, and adding your model

- `entropy-bench -baseline-only` reports `doc_bpb`: one forward pass per candidate over a few MB of the domain's text.
- Text from the task's domain, collected after the candidates shipped.

Public leaderboard, document tracks: **arXiv maths** (CC0 corpus in the repo; anyone can run it and submit) and **news** (maintainers run requested models), 17 models each. Submissions are checked against the track's corpus hash, context, stride and scored bytes.

Score the candidates on your own documents.

Questions?

Backup: the adaptation results, benchmark scores, contamination breakdown, longer-context transfer.

Backup: the adaptation results as a table

Lower BPB is better. Hand-picked recipe, 250 steps, news corpus.

Model	Architecture	Params	Zero-shot	Adapted	Reduction
Gemma-4-31B	Dense Transformer	31B	0.752	**0.624**	17.0%
Gemma-4-12B	Dense Transformer	12B	0.866	0.653	24.5%
Qwen-3.5-35B-MoE	Mixture of Experts	35B	0.707	0.678	4.1%
Ministral-3-14B	Dense Transformer	14B	0.718	0.679	5.4%
Qwen-3.5-9B	Dense Transformer	9B	0.740	0.708	4.4%
Qwen-2.5-7B	Dense Transformer	7B	0.758	0.717	5.4%
Qwen-3.5-4B	Dense Transformer	4B	0.782	0.749	4.2%
Llama-3.2-1B	Dense Transformer	1.2B	0.811	0.771	4.9%
Qwen-2.5-1.5B	Dense Transformer	1.5B	0.833	0.797	4.3%
LiquidAI-LFM2.5	Liquid Neural Net	1.2B	1.340	0.852	**36.4%**
Qwen-2.5-0.5B	Dense Transformer	0.5B	0.935	0.901	3.6%

Every value reproduces from the marker-masked fixed-protocol files in `results/domain_transfer/news__*.json`.

Backup: full benchmark scores

Eleven-model cohort, one host, one build

Model	Adapted BPB	MMLU-Pro	HellaSwag	GSM8K
Gemma-4-31B	**0.624 (1)**	57.1 (2)	**87.9 (1)**	85.0 (3)
Gemma-4-12B	0.653 (2)	44.8 (7)	84.0 (3)	64.8 (7)
Qwen-3.5-35B-MoE	0.678 (3)	**61.4 (1)**	85.6 (2)	**86.6 (1)**
Ministral-3-14B	0.679 (4)	52.8 (4)	82.9 (4)	80.8 (6)
Qwen-3.5-9B	0.708 (5)	55.6 (3)	81.6 (5)	85.8 (2)
Qwen-2.5-7B	0.717 (6)	47.2 (6)	80.3 (6)	81.3 (4)
Qwen-3.5-4B	0.749 (7)	52.4 (5)	77.2 (7)	81.0 (5)
Llama-3.2-1B	0.771 (8)	11.2 (11)	65.8 (9)	6.4 (11)
Qwen-2.5-1.5B	0.797 (9)	25.6 (8)	68.0 (8)	61.6 (8)
LiquidAI-LFM2.5	0.852 (10)	16.4 (9)	60.9 (10)	54.1 (9)
Qwen-2.5-0.5B	0.901 (11)	16.0 (10)	51.8 (11)	34.3 (10)

Percent, rank in parentheses; generated from the committed harness logs, the same table as the paper's. *lm-evaluation-harness* 0.4.12, bf16, greedy, native BOS; MMLU-Pro 5-shot CoT on a fixed 1,000-question subset, HellaSwag 10-shot length-normalised, GSM8K 5-shot flexible-extract.

Backup: contamination breakdown

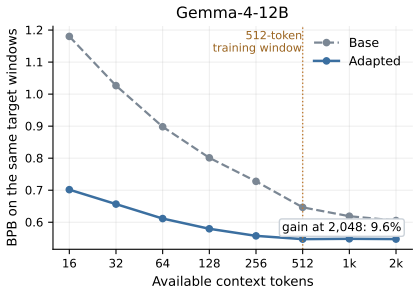
$N = 1000$ sampled sequences per model; 80 unique risk-flagged sequences removed

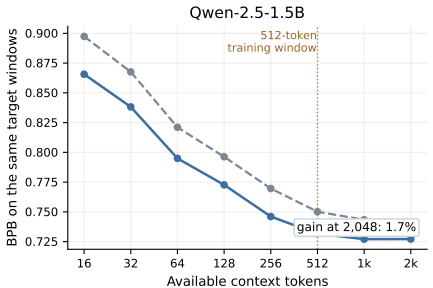
Model	Min-K++ flags	CoDeC flags	DCQ candidates	Preference	Risk flags
google-gemma-4-31B	12	6	20	100%	20
google-gemma-4-12B	11	6	11	100%	11
mistralai-Ministral-3-14B	12	8	13	100%	13
Qwen-Qwen3.5-35B-MoE	6	8	11	100%	11
Qwen-Qwen3.5-9B	12	6	17	100%	17
Qwen-Qwen2.5-7B	10	6	14	100%	14
Qwen-Qwen3.5-4B	9	6	12	100%	12
meta-llama-Llama-3.2-1B	17	0	28	100%	28
Qwen-Qwen2.5-1.5B	14	4	17	100%	17
LiquidAI-LFM2.5	12	8	23	96%	22
Qwen-Qwen2.5-0.5B	8	2	13	100%	13

These are conservative quarantine signals on a preselected subset, not verified memorization rates. Exact overlap and model-likelihood signals must be interpreted separately.

Backup: 512-token adaptation transfers to longer contexts

Same held-out targets; only the available preceding context changes





Gemma base BPB keeps recovering after 512: **0.6465** → **0.6054** at 2,048 tokens.

Adapted Gemma stays at **0.547**; its advantage remains **9.6%** at four times the training context.

The adapter does not require the inference sequence to stop at 512 tokens.